FACULTY OF ENGINEERING
CHULALONGKORN UNIVERSITY
2110201 Computer Engineering Mathematics
Year II, Second Semester, Final Examination, May 15, 2023, 13:00-16:00

Nae……Student ID…… CR58 ……

Remarks

1. **There are 21 questions in this exam paper, with the total number of 10 pages including this page.**
2. **The exam duration is 180 minutes.**
3. Write your Student ID, full name in the space provided on the top of every page.
4. No documents or calculators are allowed inside the exam room except for non-programmable and non-graphic calculators.
5. Borrowing is not allowed unless it is supervised by the proctor.
6. You must not bring any part of this exam paper outside. The exam paper is a government's property. Violators will be prosecuted under a criminal court.
7. Student who wishes to leave the exam room before the end of the exam period, must raise his/her hand and ask for permission before leaving the room. Student must leave the room in the orderly manner.
8. Once the time is expired, student must stop writing and must remain seated quietly until the proctors collect all the exam papers or given exam booklets. Only then, the students will be allowed to leave the room in the orderly manner.
9. Any student who does not obey the regulations listed above will receive punishment under the Faculty of Engineering Official Announcement on January 6, 2003 regarding the exam regulations.
   a) **With implicit evidence or showing intention for cheating, student will receive an F in that subject and will receive an academic suspension for 1 semester.**
   b) **With explicit evidence for cheating, student will receive an F in that subject and will receive an academic suspension for 1 year.**

ห้ามนิสิตพกโทรศัพท์และอุปกรณ์สื่อสารไว้กับตัวระหว่างสอบ หากตรวจพบจะถือว่านิสิตกระทำผิดเกี่ยวกับการสอบ อาจต้องพ้นสภาพการเป็นนิสิต หรือ ให้ได้รับ F และอาจพิจารณาให้ถอนรายวิชาอื่นทั้งหมดที่ลงทะเบียนไว้ในภาคการศึกษานี้

*** ร่วมรณรงค์การไม่กระทำผิดและไม่ทุจริตการสอบที่คณะวิศวกรรมศาสตร์ ***

ข้าพเจ้ายอมรับในข้อกำหนดที่กล่าวมานี้ ข้าพเจ้าเป็นผู้ทำข้อสอบนี้ด้วยตนเองโดยมิได้รับการช่วยเหลือ หรือให้ความช่วยเหลือ ในการทำข้อสอบนี้

1. Pencil is allowed.
2. Write your answer in the provided space only. If the space is not enough, you must use the back of the same page only.

ลงชื่อนิสิต……

วันที่……

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Part I: Basic Skill 3 & 4, provide detailed answers.

1. (4 points) Find the null space of the following matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 5 & 1 & 1 \\ 0 & 0 & 0 & 0 & 8 & 2 & 4 \\ 0 & 0 & 0 & 0 & 3 & 3 & 1 \\ 0 & 0 & 0 & 0 & 7 & 2 & 3 \end{bmatrix}$$

2. (2 points) Let $S = \mathrm{span}(\{v_1, v_2\})$ where $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. Find $S^{\perp}$.

3. (4 points) Let A be a 3×4 matrix such that

$$\text{RRE}(A) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \text{RRE}(A^T) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Find a basis for the null space, column space, row space and left null space of A.
What is the rank and nullity of A?

4. (3 points) Find the orthonormal basis by applying Gram-Schmidt algorithm to the given vectors in the given order, i.e., the direction of v_1 should stay the same.
$v_1 = (1,1,1)$, $v_2 = (1,1,0)$, $v_3 = (1,0,0)$

5. (6 points) Find all eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.

6. (6 points) An SVD of matrix A is given by

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & 0.5 \end{bmatrix}^T$$

Matrix A has _____ rows and _____ columns.

Rank of A = _____ Nullity of A = ____

From the SVD, write down a basis of each of the following spaces

$row(A)$

$col(A)$

$nul(A)$

$nul(A^T)$

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Part II (1 point per question): Provide short answer only.

1. A is a matrix with 3 rows and 2 columns while B is a matrix with 2 rows and 3 columns. What is the minimum and maximum rank of AB

   a. Minimum rank is_____

   b. Maximum rank is_____

2. Let A be a 3×3 orthogonal matrix but A is not the identity matrix. Find A such that $A = A^{-1}$

3. Find at least 2 matrices such that each matrix has exactly two orthonormal basis for its column space.

4. Find a 2×2 change-of-basis matrix that also represents a distortion-free mapping. The matrix must contain only number 1, 0, or -1.

5. Let h be a linear function from $\mathbb{R}^2 \to \mathbb{R}^2$ represented by A. Let D be a basis of $\mathbb{R}^2$ and D is not the standard basis. Find any basis D that makes $A = Rep_{D,D}(h)$

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6. Suppose that A is a diagonalizable matrix with characteristic polynomial $\lambda^2(\lambda-3)(\lambda+2)^3(\lambda-4)^3$.
   a. The size of the matrix A is _____
   b. The dimension of E_4, the eigenspace corresponding to eigenvalue $\lambda = 4$, is _____
   c. The rank of A is _____

7. Find the pseudoinverse of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}$.

8. The least square line of the set of points: $(1,1), (2,2), (3,3), (4,5), (5,k)$ is the line $y = 1.5x - 0.9$. Find k.

9. An SVD of matrix A is given by $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & 0.5 \end{bmatrix}^T$. How many different SVDs of matrix A are there?

10. Let $A = \begin{bmatrix} 0.1 & 2 & 4 \\ 0 & 0.2 & 3 \\ 0 & 0 & 0.3 \end{bmatrix}$. Find $\lim_{n\to\infty} A^n$.

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Part III: Show how to solve in detail (6 points each).

1. Let $A = \begin{bmatrix} 3 & -3 & x \\ -1 & 1 & y \end{bmatrix}$ and let $S = \{\begin{bmatrix} x \\ y \end{bmatrix} |$ the value of x and y make A have the dimension of $row(A)$ as 2$\}$. Calculate the set S and prove or disprove that S is a subspace.

2. Let $f: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear mapping, let $G = \mathrm{Rep}_{B,B}(f)$ where B is a basis of $\mathbb{R}^2$. G is also the change of basis matrix $\mathrm{Rep}_{B,E}(\mathrm{id})$ where E is the standard basis and id is the identity function. Let $h: \mathbb{R}^2 \to \mathbb{R}^2$ be a distortion-free mapping such that $G = \begin{bmatrix} h(\begin{bmatrix} 1 \\ 0 \end{bmatrix}) & h(\begin{bmatrix} 0 \\ 1 \end{bmatrix}) \end{bmatrix}$. We know that $G \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$. Calculate the value of $f(\begin{bmatrix} 1 \\ 0 \end{bmatrix})$ and $f(\begin{bmatrix} 0 \\ 1 \end{bmatrix})$.

3. Let U and V be a subspace in $\mathbb{R}^4$. Suppose that $\dim U = \dim V = 3$. Show that there exist two linearly independent vectors in $U \cap V$, i.e., show that there exist a linearly independent set $S \subseteq (U \cap V)$ such that S contains two vectors.

4. Let A and B be $n \times n$ matrices with a common eigenvector **x**. Show that $AB - BA$ is singular.

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5. Find all the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 2 & 2 & 2 & 2 \\ 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 2 & 2 & 2 & 1 & 2 \\ 2 & 2 & 2 & 2 & 2 & 1 \end{bmatrix}$$